Chinmay Nivsarkar

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EXPERIENCE

SIEMENS HEALTHINEERS, Princeton, NJ
Research Intern - Large Language Models Jan 2024 - May 2024

- Implemented a proof-of-concept solution for extracting relevant cases from large datasets to generate subsets for training and evaluating Machine Learning and Deep Learning algorithms, using Semantic Search as the backbone.
- Created data ingestion and management workflows for internal monitoring tools, improving turnaround times for existing processes by 25%.
- Contributed to the maintenance of legacy applications across various languages and tools and performed new feature enhancements.

SHELL INDIA MARKETS PVT. LTD, Bengaluru, India
Data Engineer Jul 2018 - Jul 2022

- Generated $40M in combined cost savings and new value realization by executing key deliverables in Azure migration and MVP deployments, employing creative solutions to overcome technological limitations.
- Architected the Databricks development pattern for the platform's migration to Azure, developing a custom framework using Scala and Spark SQL to streamline coding efforts and generate a generic code template for 99% of use cases.
- Spearheaded technical upgrades for the Azure platform, planning and managing breaking change mitigation to reduce objects requiring refactoring by 80%.
- Crafted a multitude of Azure Data Factory pipelines, implementing novel approaches to combat limitations from existing connectors.
- Drove weekly technical discussions, leading a team of over 30 developers, showcasing new developments, addressing challenges, and collaborating on technical designs for upcoming use cases.
- Executed database administration tasks in non-production environments and led quarterly maintenance activities. Contributed scripts that significantly reduced the need for constant monitoring.
- Delivered various reporting solutions for global assets, including a notable custom commenting feature on our Spotfire dashboards backed by SAP HANA, saving over 100 hours of manual work monthly.
- Designed and developed a custom data loading library in Python aimed at replacing Alteryx workflows for CSV loads in our SAP HANA platform. The library supports extended quality-of-life features such as customized column mappings, custom data-type conversions, and email notifications.

PROJECTS

***Anomaly Detection in Network Traffic* -** Python, Tensorflow, sklearn Dec 2023

- Applied machine learning models to detect ARP spoofing attacks, enhancing network security by preventing man-in-the-middle attacks and safeguarding data integrity.
- Leveraged a comprehensive dataset featuring network traffic metrics (e.g., packet size, packet rate, inter-packet delays), analyzed through traditional ML models, achieving high accuracy Random Forest (99.99%) and GRU (99.76%).

***Deepfake Detection* -** Python, PyTorch, NumPy, GANs May 2023

- Investigated various methods for detecting Deepfakes, leveraging GANs and comparing standard detection techniques with a novel approach utilizing a fine-tuned discriminator.
- Utilized the Flickr-Faces-HQ dataset to train a custom GAN for generating Deepfakes, then applied and compared different detection models, including CNNs and vision transformers.

***Data Serving for Advanced Analytics* -** Azure Data Lake Storage, Databricks, Spark, Scala, MSSQL, PowerBI Jun 2022

- Leveraged newly available Spark 3.0 features and the Databricks SQL Endpoints to create a framework that would enable direct consumption from our Datalake into PowerBI while applying row-level security to the data.
- Engineered a scalable and cost-transparent solution for consumers, enabling data consumption use cases such as machine learning requiring large volumes (1M+ rows) that were not well supported by a reporting database.

***No-Development Data Load Framework* -** Azure Data Factory, Databricks, Spark, Scala, MSSQL Apr 2021

- Extended the Databricks development patterns into a fully automated solution, enabling data loads using only configuration tables and a single Databricks notebook that could be seamlessly integrated into pre-existing Azure Data Factory pipelines.
- Reduced development effort to virtually zero for over 40% of use cases while supporting features such as custom joins, calculated columns, upserts, schema evolution, and more–saving 50-100 hours of effort per new deployment.

EDUCATION

NEW YORK UNIVERSITY, New York, NY
Master of Science in Computer Engineering May 2024

- Graduate Assistant for the High-Performance Machine Learning Operating and Systems courses.

MANIPAL INSTITUTE OF TECHNOLOGY, Manipal, India
Bachelor of Technology in Computers and Communication Engineering May 2018

SKILLS AND COURSEWORK

Languages: Python, Scala, Java, C#, C, Shell Scripting | **Databases:** MSSQL, SAP HANA, MongoDB, PostgreSQL
Others: Azure Data Factory, Databricks, Spark, NoSQL, Git, Power BI, Linux, High-Performance Computing, pandas, NumPy, Tensorflow, Pytorch, Algorithms, REST APIs
Coursework: Machine Learning, Deep Learning, High-Performance Machine Learning, Operating Systems, Data Engineering